INCH-POUND
MIL-DTL-32800
26 September 2025

DETAIL SPECIFICATION

CAP, NON-NUCLEAR, BIOLOGICAL, OR CHEMICAL (NBC), WATER CANTEEN, FIELD, 1 QUART AND 2 QUART CANTEENS

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers a plastic non-NBC canteen water cap for the 1 quart and 2 quart water canteens.

2. APPLICABLE PUBLICATIONS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-43103 Canteen, Water, Plastic, With Screw Cap
MIL-C-43603 Canteen, Water, Collapsible, 2-Quart Capacity

(Copies of this document are available online at https://quicksearch.dla.mil)

Comments, suggestions, or questions on this document should be sent to trpsptctstandardiz@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

AMSC N/A FSC 8465

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY, DEVCOM – SOLDIER CENTER

2-6-1514	-	CAP, NON-NBC, WATER CANTEEN, 2 QUART CANTEEN
2-6-1515	-	STRAP, NON-NBC, WATER CANTEEN, 2 QUART CANTEEN
8-1-33	-	CANTEEN, WATER, PLASTIC, ASSEMBLY AND DETAILS
8-2-228	-	CANTEEN, WATER, COLLAPSIBLE, 2-QUART
136-5-520-1	-	TESTER, AIR LEAKAGE, CANTEEN CAP, Q178
136-300-261	-	INSTRUCTION MANUAL FOR THE INSTALLATION OPERATION AND MAINTENANCE OF TESTER, AIR LEAKAGE, CANTEEN CAP, Q178

(Copies of drawings are available from the U.S. Army, DEVCOM Soldier Center, ATTN: FCDD-SCD-PEC, 10 General Greene Avenue, Natick, MA 01760-5019).

(Comments, suggestions, or questions on this document should be addressed to: U.S. Army, DEVCOM – Soldier Center, ATTN: FCDD-SCD-PEC, 10 General Greene Avenue, Natick, MA 01760-5019).

OTHER GOVERNMENT DOCUMENTS AND PUBLICATIONS

CODE OF FEDERAL REGULATIONS

FOOD AND DRUG ADMINISTRATION, DEPARTMENT OF HEALTH AND HUMAN SERVICES

Title 21 CFR Part 177 Indirect Food Additives: Polymers

(Copies of these documents are available online at https://www.ecfr.gov).

(Copies of specifications, standards, drawings and publications required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles

(Copies of these documents are available online at https://www.aatcc.org).

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org).

ASTM INTERNATIONAL

ASTM D638 - Standard Test Method for Tensile Properties of Plastics
ASTM D790 - Standard Test Method for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
ASTM D1238 - Standard Test Method for Flow Rates of Thermoplastics by Extrusion Plastometer
ASTM D2240 - Standard Test Method for Rubber Property - Durometer Hardness

(Copies of these documents are available online at https://www.astm.org).

SAE INTERNATIONAL

AMS-STD-595 – Colors used in Government Procurement

(Copies of this document are available online at https://www.sae.org/).

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3.1, 6.2, and 6.3).

3.2 Materials and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document.

3.2.1 Plastic. The plastic material for the cap shall be a High-Performance Acetal Homopolymer (see 6.4) conforming to the requirements specified in Table I when tested as

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specified in 4.4. The materials used in the manufacture of the cap plus other materials used in the fabrication of the plastic shall conform to the requirements of Title 21 CFR Part 177 – Indirect Food Additives: Polymers as specified by the Food and Drug Administration, Department of Health and Human Services. (see 4.4.1.1).

TABLE I. Material requirements.

Characteristic	Requirement (Nominal)
Melt flow rate, g/10 min	7
Tensile strength, psi	5,200
Elongation at yield, $g.cm^3$	1.42
Hardness (Rockwell Hardness R)	120
Flexural modulus, psi	450,000
Color	OD Green

3.2.1.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.2.1.2 Toxicity. The material used in the manufacture of the cap plus other materials used in the fabrication of the plastic shall conform to the requirements of 21 CFR Part 177 – Indirect Food Additives: Polymers (see 4.4.1.1). Unless otherwise specified in the contract or procurement documents, the finished Cap shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.4.1.1.

3.3 Design and construction. The design and construction of the cap assembly shall be as specified herein and shown on included drawings.

3.4 Molded cap requirements.

3.4.1 Molding method. The molded cap shall be injection molded from the plastic material specified in 3.2.1.

3.4.2 Color. The color of the cap shall be a good match to OD Green color chip 34078 lusterless, of AMS-STD-595.

3.4.2.1 Visual shade matching. The color and appearance of the Cap shall approximate the color in 3.4.2 when tested as specified in 4.5.4.

3.4.3 Torque strength. The molded cap shall not slip, disengage, or crack when tested as specified in 4.5.3.

3.5 Assembled cap requirements.

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3.5.1 Leakage resistance. The cap assembly shall not leak when tested as specified in 4.4.5.

3.5.1.1 Compatibility. The cap shall fit the Canteen, Water, Plastic, With Screw Cap and Canteen, Water, Collapsible, 2 quart capacity as specified in MIL-C-43103, MIL-C-43603 and drawings 8-1-33, 8-2-228 when tested as specified in 4.5.1.

3.6 Workmanship. The cap shall be free from fractures, splits, punctures, tears, dents, deteriorations, or malformations. The exterior surfaces of the cap shall have a smooth and lusterless finish throughout and shall be free of dirt, dust, grease and foreign matter. The finish shall be produced from a die which has cavity surfaces finished in vapor blast, water hone or similar satin finish and not by the application of any protective coating, lacquers or other materials.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1)
b. Conformance inspection (see 4.3.2)

4.2 Inspection conditions. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.3 Types of inspections.

4.3.1 First article inspection. The first article, submitted in accordance with 3.1 shall be inspected for appearance, color, finished measurements included in drawings 2-6-1514 and 2-6-1515, examination for defects in 4.4.3, Table IV and tested as specified in 4.4.4, 4.4.5 and 4.5 through 4.5.4. The presence of any defect or failure to pass any test shall be because of rejection of the first article.

4.3.2 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.4 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits as specified in the contract and/or procurement documents, except where otherwise indicated.

4.3.3 Certificates of compliance. When certificates of compliance (see 6.5) are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.3.4 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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4.4 Material testing.

4.4.1 Material certification. The material for the cap, (see 3.2.1) shall be accepted based on a contractor's certificate of compliance with the characteristics listed in Table II.

TABLE II. Plastic raw material tests.

Characteristic	Requirement	Test method
Melt flow rate	Table I	ASTM D1238, Procedure A, condition E
Tensile strength	Table I	ASTM D638, Type IV specimens
Elongation at yield	Table I	ASTM D638, Type IV specimens
Hardness (Rockwell Hardness R)	Table I	ASTM D2240
Flexural modulus	Table I	ASTM D790, Procedure B
Color	Table I and 3.4.2	Visual (see 4.5.4)

4.4.1.1 Toxicity test. A certificate of compliance (see 6.5) will be acceptable as evidence of conformance to the toxicity requirements specified in 3.2.1.2. If dermal toxicity testing is required, it shall be conducted using Title 40, Protection of Environment, Code of Federal Regulations current edition and applicable sections, which are also consistent with the following OECD tests:

Skin Sensitization 406 and Dermal Irritation 404 (https://www.oecd-ilibrary.org/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects_20745788?page=1).

Dermal Sensitization (https://www.law.cornell.edu/cfr/text/40/798.4100).

Dermal Toxicity (https://www.law.cornell.edu/cfr/text/40/798.2250).

4.4.1.2 Cap components (other than cap shell) certification. A certificate of compliance (see 6.5) will be acceptable as evidence that the gasket conforms to the material and dimensional requirements specified on drawing 2-6-1514. The strap shall meet the requirements as specified in drawing 2-6-1515.

4.4.2 In-process inspection.

4.4.2.1 Mold approval. Prior to use in regular production, each mold shall be used to produce an initial production lot of cap shells, and one (1) cap shell shall be randomly selected from that lot. The cap shell shall be examined for conformance to all dimensions specified on drawing 2-6-1514. If the cap is found to conform to all dimensional requirements, the related mold shall be approved. If the cap shell fails to conform to any dimensional requirement, an additional five (5) cap shells shall be randomly selected from the initial production lot and examined for all dimensions. If any additional cap shell fails to conform to any dimension, the related mold and initial production lot shall be rejected. When any mold is repaired or altered in any way, mold reapproval in accordance with the above procedure is required.

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4.4.2.2 In-process testing of cap shells. Prior to beginning regular production, four (4) cap shells shall be selected at random from the initial production lot and tested as specified in Table III. Any test failure shall be cause for rejection of the lot. Necessary changes to the materials and procedure shall be made and testing as above shall be repeated. Whenever any change in material or manufacturing process is made, the initial lot produced after the change shall be tested as above.

TABLE III. Molded cap shells in-process tests.

Characteristic	Requirement	Test method	Number of samples
Torque strength	3.4.3	4.5.3	25
Leakage test	3.5.1	4.5.2	25

4.4.3 End item visual examination. The end item shall be examined for the defects listed in Table IV. The lot size shall be expressed in units of caps. The sample unit shall be one (1) cap.

TABLE IV. End item visual defects.

Examine	Defect	Classification	
		Major	Minor
Finish	Color not as specified	101	
	Dirt, grease, oil, foreign matter, undisposed pigment, dark particles of burnt material, porosity	102	
Design, construction and workmanship general (applicable to all components and assemblies)	Any component or part missing or not specified type or size	103	
	Fractured, split, punctured, torn, dented, deteriorated, or malformed	104	
	Component misplaced or not in proper alignment		201
	Sharp burr, edge, or sliver		202
	Component loose		203
	Any operation omitted or not properly performed	105	
	Any functioning component that is inoperative or will not operate as intended	106	
	Any functioning component that requires abnormal force to operate		204

4.4.4 End item dimensional examination. The end item shall be examined for conformance to the dimensions annotated with an asterisk on the drawings. Any dimension not within the specified tolerance shall be classified as a defect. The lot shall be expressed in units of canteen caps. The sample unit shall be one (1) canteen cap. The inspection level and the Acquisition Quality Limits (AQL) shall be as stated in the contract.

4.4.5 End item testing. The end item shall be tested for leak resistance as specified in 4.5.2. The lot size shall be expressed in units of caps. The sample unit shall be one (1) complete cap

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assembly. The inspection level shall be as stated in the contract. Any test failure shall be cause for rejection of the lot.

4.5 Methods of inspection.

4.5.1 Compatibility. The cap shall be compatible and provide a secure seal with the 1- and 2-quart water canteens specified in MIL-C-43103, MIL-C-43603 and drawings 8-1-33, 8-2-228. The cap shall achieve a leak-tight seal when hand tightened to no more than 100 inch-pounds of torque.

4.5.2 Cap leakage test. The cap shall be connected to the Q178 air leakage tester specified in drawing 136-5-520-1 and tested according to section 7.d. of the Q178 air leakage tester manual, document 136-300-261, set to a pressure differential of 3-inches of mercury for a minimum of 5 seconds. Any leakage, as indicated by bubbles in the water in the adapter or by a change in the mercury level, shall constitute a failure. Any nonconformance to the leakage requirement of 3.5.1 shall constitute failure of the test.

4.5.3 Cap torque test. The molded cap shall be placed on a test fixture made to the dimensions of the 2 quart canteen neck and 100 inch-pounds of torque applied. Any nonconformance to the requirements of 3.4.3 shall constitute failure of the test.

4.5.4 Visual shade matching. The color and appearance of the Cap shall match the requirement of 3.4.2 when viewed using the AATCC EP 9, Option C, (see 6.6) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the Cap shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65. (See 6.6.1). The Cap shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

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6.1 Intended use. The cap is intended for use with the 1- and 2-quart military field drinking water canteens.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The specific issue of individual documents referenced (see 2.).
c. When first article is required (see 3.1).
d. When toxicity is required (see 3.2.1.2).
e. Inspection conditions (see 4.2).
g. Performance inspection with acceptance quality limits (AQL) (see 4.3.2).
h. Packaging (see 5.1).

6.3 First article. When a first article inspection is required, it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Delrin ® a product of Delrin fulfills the requirements of 3.2.1. Contact information: https://www.delrin.com/.

6.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for that specified requirement.

6.6 Visual shade matching. In 2017, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be “The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens”.

6.6.1 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75. DD1222’s will reflect which light source is used. If the light source is not indicated on the DD1222, D65 will be assumed by the Government Shade Laboratory.

6.7 Subject term (key word) listing.

Container
Cover
Drinking
Hydration
Strap

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Custodians:
Army – GL
Navy – NU
Air Force- 11

Preparing activity:
DLA-CT

(Project: 8465-2025-006)

Review activities:
Navy – MC

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